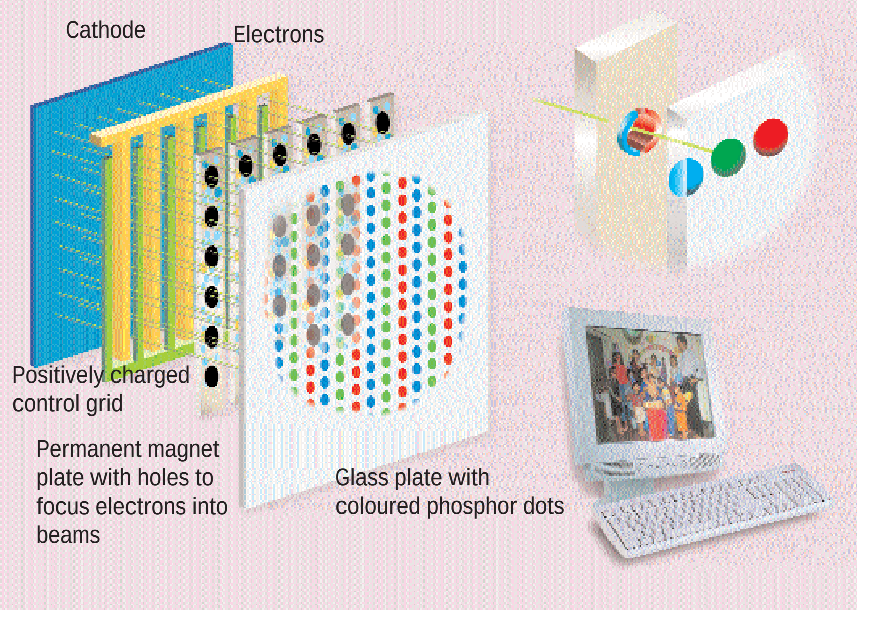
IBM's Idea of Snoop-free Viewing

Traditional CRTs (Cathode Ray Tubes) make use of vertical and horizontal deflection coils to guide an electron beam so that an image is displayed on the screen. These deflections cause the CRT to emit radiations of varying frequency.

The main problem with these radiations is that they can be picked up by equipment specially designed for this purpose.

IBM has come out with a new CRT technology that uses permanent magnets instead of deflection coils to guide the beam.

This display is made up of precision-aligned layers and is much cheaper than present day LCD displays. A flat cathode plate in the back of the display generates the stream of electrons. These electrons are focused into an electron beam by a charged control grid, much like a lens focuses light.



Now comes the major difference. A permanent magnet plate with small holes comes between the grid and the phosphorescent screen. The strong magnetic field inside each of the holes makes the electrons spin very fast, which converts the stream of electrons into focused beams. The beam then falls on the phosphorescent screen, which consists of red, green, and blue phosphor dots. As the beam is controlled by the magnets and grids, each electron hits one of the three dots and colour images are generated.

IBM claims that its new CRT technology emits only steady magnetic fields. The plus point is that the fields remain the same throughout the CRT's operation. These magnetic fields cannot be converted into data, unlike conventional CRT emissions.

Spies are not able to decipher these radiations since they are constant throughout the operation of the monitor.

for keywords, addresses, names and so on to find compromising information.

Spying—can it be prevented?

Before you start eyeing all wireless devices with extreme suspicion, lets take a step back and see how real is the threat of such snooping as far as individuals are concerned. To what extent have these technologies developed and infiltrated our lives through devices that most of us use without a second thought?

Well, the truth is that with the advent of systems such as the Global Positioning System (GPS) and biometrics, the day is not too far when you can, theoretically speaking, be tracked wherever you are, whatever you do. Your mobile phone can help pinpoint your location.

Mobile tracking is already done by police agencies as well as other law enforcement agencies. And this information could be used by, say, companies to build a bank of your tastes, likes and dislikes to better tailor their personalised marketing strategies.

But that time is still a far way off, at least in India. As of now, such technologies are primarily used for military espionage and increasingly now to sneak up on corporate rivals to defeat their strategies

But if you are still feeling paranoid, you can take some simple steps to prevent people from spying on you. For instance, you can install a personal firewall on your system to thwart attempts at cracking into it or at least make it difficult for hackers. Use encryption programs such as PGP to encrypt your mail and hard disk data. If you have to handle sensitive data, it would be advisable to avoid using mobile phones or even landlines for that matter. Use them only if there is no other alternative. In fact, avoid using wireless devices and networks when you are handling important information—at least until 802.11 has advanced to a state where it can be called secure. Also, there are several devices such as TEMPEST shielding equipment as well as shielded devices which provide you protection from spying technologies.

No technology is secure, but you can prevent it from wrecking havoc in your life by being aware and taking some precautions. So keep your cool, stay informed, and don't get bugged! d